Foundation Mathematics 1017SCG
Week 1 Workshop Solutions

Remember that the method shown in the solutions is not the only way to solve the question. There are usually multiple ways to solve the same question.

1. (a)

$$\begin{aligned} 12 - 1 - 8 &= 11 - 8 \\ &= 3 \end{aligned}$$

(b)

$$\begin{aligned} 20 - 3^2 &= 20 - 9 \\ &= 11 \end{aligned}$$

(c)

$$\begin{aligned} 4 + 10 \div 2 - 7 &= 4 + 5 - 7 \\ &= 9 - 7 \\ &= 2 \end{aligned}$$

(d)

$$\begin{aligned} 6^2 + (10 + 5) \div 3 &= 6^2 + 15 \div 3 \\ &= 36 + 15 \div 3 \\ &= 36 + 5 \\ &= 41 \end{aligned}$$

(e)

$$\begin{aligned} (1 + 5) \times (15 - 6) &= 6 \times 9 \\ &= 54 \end{aligned}$$

(f)

$$\begin{aligned} (3 + (-2)^2) \times -6 &= (3 + 4) \times -6 \\ &= 7 \times -6 \\ &= -42 \end{aligned}$$

(g)

$$\begin{aligned} -4^2 &= -(4 \times 4) \\ &= -16 \end{aligned}$$

It is important to understand the difference between the above question and the question below which looks similar.

$$(-4)^2 = (-4) \times (-4) = 16$$

(h)

$$\begin{aligned}5^2-3^2+2^2&=25-3^2+2^2\\&=25-9+2^2\\&=25-9+4\\&=16+4\\&=20\end{aligned}$$

(i)

$$\begin{aligned}60\div5\div3&=12\div3\\&=4\end{aligned}$$

(j)

$$\begin{aligned}100-2\times5\times7&=100-10\times7\\&=100-70\\&=30\end{aligned}$$

(k)

$$\begin{aligned}1.876\times10^3&=1.876\times1000\\&=1876\end{aligned}$$

2. (a)

$$\begin{aligned}4x^3\times2x^5&=4\times2\times x^3\times x^5\\&=8\times x^{3+5}\\&=8x^8\end{aligned}$$

(b)

$$\begin{aligned}9x^2y^3\times5xy^4&=9\times5\times x^2\times x^1\times y^3\times y^4\\&=45\times x^{2+1}\times y^{3+4}\\&=45x^3y^7\end{aligned}$$

(c)

$$\begin{aligned}\frac{8x^5}{10x^2}&=\frac{8}{10}\times x^{5-2}\\&=\frac{4}{5}\times x^3\\&=\frac{4x^3}{5}\end{aligned}$$

(d)

$$\begin{aligned}\frac{12x^6y^2}{16x^3y^7}&=\frac{12}{16}\times x^{6-3}\times y^{2-7}\\&=\frac{3}{4}\times x^3\times y^{-5}\\&=\frac{3x^3}{4y^5}\end{aligned}$$

(e)

$$\begin{aligned} x^2 \times (3x^2)^3 &= x^2 \times (3^1 \times x^2)^3 \\ &= x^2 \times 3^{1\times 3} \times x^{2\times 3} \\ &= x^2 \times 3^3 \times x^6 \\ &= x^2 \times 27 \times x^6 \\ &= 27 \times x^{2+6} \\ &= 27x^8 \end{aligned}$$

(f)

$$\begin{aligned} x^3y \times (2xy^4)^2 &= x^3 \times y^1 \times (2^1 \times x^1 \times y^4)^2 \\ &= x^3 \times y^1 \times 2^{1\times 2} \times x^{1\times 2} \times y^{4\times 2} \\ &= x^3 \times y^1 \times 2^2 \times x^2 \times y^8 \\ &= x^3 \times y^1 \times 4 \times x^2 \times y^8 \\ &= 4 \times x^{3+2} \times y^{1+8} \\ &= 4x^5y^9 \end{aligned}$$

(g)

$$\begin{aligned} (16x^4y^6)^{\frac{1}{2}} &= (16^1 \times x^4 \times y^6)^{\frac{1}{2}} \\ &= 16^{1\times\frac{1}{2}} \times x^{4\times\frac{1}{2}} \times y^{6\times\frac{1}{2}} \\ &= 16^{\frac{1}{2}} \times x^2 \times y^3 \\ &= \sqrt{16} \times x^2 \times y^3 \\ &= 4x^2y^3 \end{aligned}$$

(h)

$$\begin{aligned} \left(\frac{4x^2}{y^3}\right)^{-2} &= \left(\frac{4^1 \times x^2}{y^3}\right)^{-2} \\ &= \frac{4^{1\times -2} \times x^{2\times -2}}{y^{3\times -2}} \\ &= \frac{4^{-2}x^{-4}}{y^{-6}} \\ &= \frac{y^6}{4^2x^4} \\ &= \frac{y^6}{16x^4} \end{aligned}$$

3. (a) 3.1248 rounded to 1 decimal place is 3.1
   (b) 2.8942 rounded to 1 decimal place is 2.9
   (c) 12.7962 rounded to 2 decimal places is 12.80
   (d) 36.0196 rounded to 2 decimal places is 36.02
4. (a) $870,000 = 8.7 \times 10^5$
   (b) $23,400 = 2.34 \times 10^4$

(c) $4,500,000 = 4.5 \times 10^6$

(d) $0.007 = 7 \times 10^{-3}$

(e) $0.00004 = 4 \times 10^{-5}$

(f) $0.00059 = 5.9 \times 10^{-4}$

5. (a) $3.5 \times 10^3 = 3,500$

(b) $9 \times 10^6 = 9,000,000$

(c) $5.71 \times 10^2 = 571$

(d) $4 \times 10^{-3} = 0.004$

(e) $8.15 \times 10^{-5} = 0.0000815$

(f) $9.2 \times 10^{-6} = 0.0000092$

6. (a) $5.61 \times 10^4 + 3.2 \times 10^5 = 376,100$

(b) $2.8 \times 10^{-5} + 9.31 \times 10^{-4} = 0.000959$

7. (a)

$$3\pi + 5\pi = 8\pi$$

(b) Remember that when adding or subtracting fractions, you need a common denominator.

$$\begin{aligned} 2\pi - \frac{\pi}{3} &= \frac{2\pi}{1} - \frac{1\pi}{3} \\ &= \frac{6\pi}{3} - \frac{\pi}{3} \\ &= \frac{5\pi}{3} \end{aligned}$$

(c)

$$\begin{aligned} \pi + \frac{5\pi}{6} &= \frac{1\pi}{1} + \frac{5\pi}{6} \\ &= \frac{6\pi}{6} + \frac{5\pi}{6} \\ &= \frac{11\pi}{6} \end{aligned}$$

(d)

$$\begin{aligned} 2\pi - \frac{\pi}{6} &= \frac{2\pi}{1} - \frac{\pi}{6} \\ &= \frac{12\pi}{6} - \frac{\pi}{6} \\ &= \frac{11\pi}{6} \end{aligned}$$

(e)

$$\begin{aligned} \pi - \frac{\pi}{4} &= \frac{1\pi}{1} - \frac{1\pi}{4} \\ &= \frac{4\pi}{4} - \frac{1\pi}{4} \\ &= \frac{3\pi}{4} \end{aligned}$$

8. Throughout this question, the following steps have been followed.

    1. The numerators have been combined using index laws
    2. The division completed using index laws.
    3. The final answer rewritten (if necessary) to only include positive indices/powers.

    By following the above steps, intermediate steps are able to be included in these solutions. **However, you will likely find these questions much easier if you cancel before combining the fractions.**

    (a)

$$\begin{aligned}\frac{6x^2y^3}{x^4} \times \frac{x^4}{3y} &= \frac{6x^{2+4}y^3}{3x^4y}\\ &= \frac{6x^6y^3}{3x^4y^1}\\ &= \frac{6}{3} \times x^{6-4} \times y^{3-1}\\ &= 2 \times x^2 \times y^2\\ &= 2x^2y^2\end{aligned}$$

    (b)

$$\begin{aligned}\frac{10xy^6}{4x^3y^2} \times \frac{x^2y^4}{x^3} &= \frac{10x^{1+2}y^{6+4}}{4x^{3+3}y^2}\\ &= \frac{10x^3y^{10}}{4x^6y^2}\\ &= \frac{10}{4} \times x^{3-6} \times y^{10-2}\\ &= \frac{5}{2} \times x^{-3} \times y^8\\ &= \frac{5y^8}{2x^3}\end{aligned}$$

    (c)

$$\begin{aligned}\frac{10x^2y}{x^3} \div \frac{15y^5}{x^4} &= \frac{10x^2y}{x^3} \times \frac{x^4}{15y^5}\\ &= \frac{10x^{2+4}y}{15x^3y^5}\\ &= \frac{10x^6y}{15x^3y^5}\\ &= \frac{10}{15} \times x^{6-3} \times y^{1-5}\\ &= \frac{2}{3} \times x^3 \times y^{-4}\\ &= \frac{2x^3}{3y^4}\end{aligned}$$

(d)

$$\begin{aligned}
\frac{x^5z^3}{2y^4} \div \frac{x^3z^3}{4x^2y^2} &= \frac{x^5z^3}{2y^4} \times \frac{4x^2y^2}{x^3z^3} \\
&= \frac{4x^{5+2}y^2z^3}{2x^3y^4z^3} \\
&= \frac{4x^7y^2z^3}{2x^3z^3} \\
&= \frac{4}{2} \times x^{7-3} \times y^{2-4} \times z^{3-3} \\
&= 2 \times x^4 \times y^{-2} \times z^0 \\
&= \frac{2x^4}{y^2} \qquad\qquad \text{(Remember } z^0 = 1\text{)}
\end{aligned}$$

9. (a) At the end of Foundation Mathematics, most students will be awarded either a non-graded pass (NGP) or a non-graded fail (NGF). Technically there are a few other grades, such as withdraw (W) or withdraw fail (WF), but most students will receive either NGP or NGF. **Note that as Foundation Mathematics is a non-graded course, grades such as Credit (5) and High Distinction (7) are not possible.**

   (b) To be eligible for a non-graded pass at the end of the trimester, you must achieve a minimum of 65% overall for the course (applying the appropriate weightings for each assessment item).